Remark 3.15. By [19, Remark 2.28], if a definable function between two sets in Γ is a vrv-contraction then it is piecewise $\mathbb{Z}$-affine (with constant terms of the form $\mathrm{vrv}(t)$, where $t \in \mathrm{RV}$ is definable, that is, $t \in \mathrm{RV}(\mathbb{S})$). Moreover, a definable bijection between two objects of $\Gamma[k]$ is a $\Gamma[k]$-morphism if and only if it is definably a piecewise $\mathrm{GL}_k(\mathbb{Z})$-transformation. The "if" direction is clear. For the "only if" direction, see [8, Lemma 10.1] or [19, Lemma 2.29].

Resolved referee R050 [Clarification] – Piecewise integral-affine behavior.
Original source line 1289. The first sentence needs "piecewise $\mathbb{Z}$-affine," not globally "$\mathbb{Z}$-linear." For example, $\gamma \mapsto |\gamma|$ is the vrv-contraction of the definable map $t \mapsto t$ for $\mathrm{vrv}(t) \geq 0$, $t \mapsto t^{-1}$ otherwise. Its two integral-linear formulas differ. Retain the specified restriction on the translation constants.

Author resolution. The author approves the piecewise $\mathbb{Z}$-affine formulation. The restriction on translation constants and the following piecewise $\mathrm{GL}_k(\mathbb{Z})$ characterization are retained.

Definition 3.16 (Γ-categories with volume forms)**.** An object of the category $\mu\Gamma[k]$ is a definable pair (I, ω), where $I \in \Gamma[k]$ and $\omega : I \longrightarrow \Gamma$ is a function. A $\mu\Gamma[k]$-morphism between two objects (I, ω), (J, σ) is a $\Gamma[k]$-morphism $F : I \longrightarrow J$ such that for all $\alpha \in I$,

$$\omega(\alpha) = \sigma(F(\alpha)) + \mathrm{Jcb}_\Gamma F(\alpha).$$

The category $\mu\Gamma^{\mathrm{fin}}[k]$ is the obvious full subcategory of $\mu\Gamma[k]$.

Notation 3.17. For each $(U, f, \omega) \in \mu\mathrm{RV}[k]$, write $\omega_f : U \longrightarrow \Gamma$ for the function given by $u \longmapsto \Sigma(\mathrm{vrv} \circ f)(u) + \omega(u)$. Similarly, for $(I, \sigma) \in \mu\Gamma[k]$, write $\sigma_I : I \longrightarrow \Gamma$ for the function given by $\gamma \longmapsto \Sigma\gamma + \sigma(\gamma)$.

There is a natural map $\Gamma[*] \longrightarrow \mathrm{RV}[*]$ given by $I \longmapsto (I^\sharp, \mathrm{id})$ (see Notation 2.7). This map induces a commutative diagram in the category of graded semirings:

$$\begin{array}{ccc} \mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RES}[*] \\ \downarrow & & \downarrow \\ \mathbf{K}_+\,\Gamma[*] & \rightarrowtail & \mathbf{K}_+\,\mathrm{RV}[*] \end{array}$$

where all the arrows are monomorphisms. The map

$$\mathbf{K}_+\,\mathrm{RES}[*] \times \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*]$$

determined by the assignment

$$([(U, f)], [I]) \longmapsto [(U \times I^\sharp, f \times \mathrm{id})] \tag{3.3}$$

is well-defined and is clearly $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-bilinear. Hence it induces a $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-linear map

$$\Psi : \mathbf{K}_+\,\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\Gamma[*] \longrightarrow \mathbf{K}_+\,\mathrm{RV}[*], \tag{3.4}$$

which is a homomorphism of graded semirings. By the universal mapping property, groupifying a tensor product in the category of $\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]$-semimodules is, up to isomorphism, the same as taking the corresponding tensor product in the category of $\mathbf{K}\,\Gamma^{\mathrm{fin}}[*]$-modules; the groupification of Ψ is still denoted by Ψ. Similarly, there is a $\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]$-linear map

$$\mu\Psi : \mathbf{K}_+\,\mu\mathrm{RES}[*] \otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]} \mathbf{K}_+\,\mu\Gamma[*] \longrightarrow \mathbf{K}_+\,\mu\mathrm{RV}[*]. \tag{3.5}$$

We shall abbreviate $\otimes_{\mathbf{K}_+\,\Gamma^{\mathrm{fin}}[*]}$, $\otimes_{\mathbf{K}_+\,\mu\Gamma^{\mathrm{fin}}[*]}$ both as "$\otimes$" below when no confusion can arise.

Proposition 3.18 ([8, Corollary 10.3, Proposition 10.10(1)])**.** *Ψ and $\mu\Psi$ are both isomorphisms of graded semirings.*

Formalization review T001: finite coefficients and the fixed ground. The injectivity assertion above needs review under the present parameter and Γ-morphism conventions. The following obstruction already occurs for a VF-generated henselian ground with nontrivial value group. Take $\mathbb{S} = \mathbb{C}((t))$, normalize $\mathrm{val}(t) = 1$, and put

$$\delta = \tfrac{1}{5}, \qquad R = \{r \in \mathrm{RV} : r^5 = \mathrm{rv}(t)\}, \qquad I = (0,\infty), \quad J = (\delta,\infty).$$

Here R is a five-element object of RES[0] and every $r \in R$ has value δ. The original-ground definable bijection

$$R \times I^\sharp \longrightarrow R \times J^\sharp, \qquad (r,x) \longmapsto (r,rx)$$

has inverse $(r,y) \mapsto (r,r^{-1}y)$. Thus $\Psi([R]_0 \otimes [I]_1) = \Psi([R]_0 \otimes [J]_1)$. The two tensors are nevertheless distinct with the stated balancing ring.

A detector for their difference. All definable finite endpoints in Γ are rational. Define $F : \mathbb{Q} \longrightarrow \mathbb{Z}$ by $F(a) = 1$ on $\frac{1}{5} + \mathbb{Z}$, $F(a) = -1$ on $-\frac{1}{5} + \mathbb{Z}$, and $F(a) = 0$ otherwise; set $F(\pm\infty) = 0$. It is odd and $\mathbb{Z}$-periodic. Give each interval with endpoints $a < b$ weight $F(a) - F(b)$, independently of endpoint inclusion, and each point weight zero. Add the weights over finite disjoint unions to obtain ν. Cutting an interval at c preserves its weight because $(F(a) - F(c)) + (F(c) - F(b)) = F(a) - F(b)$. Translations by integers and reflections $a \mapsto -a + z$, $z \in \mathbb{Z}$, also preserve the weight. Consequently ν is an additive invariant of $\Gamma[1]$: these are exactly the permitted affine pieces in this ground by Remark 3.15. In particular, $\nu(I) = 0$, $\nu(J) = 1$, and ν vanishes on finite objects.

Let $D = \mathbb{Z}[\varepsilon]/(\varepsilon^2)$, graded by $\deg \varepsilon = 1$. There are graded semiring homomorphisms

$$\mathbf{K}_+ \,\mathrm{RES}[*] \longrightarrow D, \quad [X]_0 \longmapsto |X|, \quad [X]_k \longmapsto 0 \ (k > 0),$$

$$\mathbf{K}_+ \,\Gamma[*] \longrightarrow D, \quad [X]_0 \longmapsto |X|, \quad [X]_1 \longmapsto \nu(X)\varepsilon, \quad [X]_k \longmapsto 0 \ (k \geq 2).$$

They agree on $\mathbf{K}_+ \,\Gamma^{\mathrm{fin}}[*]$, so induce a homomorphism from the tensor source of Ψ. It sends the displayed tensors to 0 and 5ε, respectively. This proves their inequality, also after groupification. The finite-Γ balancing relations therefore do not suffice to descend this coefficient-dependent translation.

HK convention audit. I checked the parameter and algebraic base-change discussion in HK §§ 1–3, the effective/resolved distinctions and base-change summary, the grading and localization constructions in § 9, the full tensor proof in § 10, and the henselian-ground discussion in § 12. I also compared the critical passages in arXiv versions 2 and 3 and readable cached text of Oxford's accepted-manuscript copy. No convention was found that removes the obstruction for this unlocalized graded tensor. Naming algebraic parameters in § 2.1.4 is an explicit change of theory, not a blanket replacement of the fixed ground throughout the comparison.

The localization deserves particular care. Put $\tau = [\{0\}]_1 \in \mathbf{K}_+ \,\Gamma^{\mathrm{fin}}[1]$, and let a, b denote the two source tensors above. In fact $\tau a = \tau b$: the residue bijection $R \times 0^\sharp \cong R \times \delta^\sharp$ and the integral unimodular shear $(\alpha,\gamma) \mapsto (\alpha, \gamma + \alpha)$ on $\{\delta\} \times I$ give this equality by balancing. Thus inverting τ removes this example. But τ has degree one and is not the multiplicative identity. HK's localized constructions in § 9 are separate from Corollary 10.3; that corollary does not invert τ. The detector above sends τ to zero and still distinguishes a and b.

The exact point requiring justification remains the equality of tensor terms after Claim 4 of Proposition 10.2 (printed page 92 in arXiv v3). A definable Γ factor map must have a lift over the original ground; a lift obtained after fixing $r \in R$ does not by itself establish that. Interpreting all divisible-hull translations as allowed would not fix this: the arrow $\{0\} \longrightarrow \{1/5\}$ would then need an RV lift, whereas $0^\sharp$ has a definable point and $(1/5)^\sharp$ has none over this ground. No verified replacement theorem or change of conventions is proposed. The working manuscript is unchanged. The concrete model and the complete detector have not yet been constructed in Lean.

Note that, however, [8, Proposition 10.10(2)] does not hold. One of the main motivations of this paper is to construct an alternative to be used in applications.